Minh-Nhat Nguyen

Undergraduate AI Researcher

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Summary

A Computer Science undergraduate with a solid mathematical foundation and hands-on research in Computer Vision and Generative Models (Diffusion Models, Flow Matching). Experienced in developing training-free generative methods, multimodal video retrieval, and architectural document understanding pipelines.

Education

Bachelor of Information Technology — **University of Science, VNUHCM**, Information Technology (Computer Science) — Ho Chi Minh City, Vietnam, 2025 – 2029
- GPA: 3.74/4.0 (8.75/10.0), All core mathematics and CS courses: 4.0/4.0.
- Expected graduation: 2029
- Advanced Online Coursework: MIT 6.S184: Generative AI with Stochastic Differential Equations.

High School Diploma — **Le Quy Don Gifted High School**, Mathematics — Da Nang, Vietnam, 2022 – 2025
- Specialized in Advanced Mathematics.

Experience

SGF Group Pty Ltd, AI Research Intern — Ho Chi Minh City, Vietnam, Oct 2026 – present, 2 months
- Engineered an automated architectural drawing recognition and component counting system from vector PDF floorplans, accurately detecting doors and structural building elements.
- Developed preprocessing and geometric post-processing algorithms integrated with Vision Foundation Models (VFMs) to significantly improve recognition precision.
- Implemented semi-supervised fine-tuning tailored to heterogeneous CAD datasets and deployed the complete inference recognition pipeline.

University of Science, VNUHCM, Undergraduate AI Researcher — Ho Chi Minh City, Vietnam, Mar 2026 – present, 9 months
- Developed training-free methods for style transfer using Latent Diffusion Models and Flow Matching, introducing latent patch-shuffling to eliminate content leakage and pixel-space grid artifacts.
- Optimized generative algorithms by manipulating sampling trajectories, guidance terms, and cross-attention maps to balance stylistic and structural fidelity.

Publications

OrthoStyle: Manifold-Preserving Training-Free Style Transfer via Score-Orthogonal Guidance — Oct 2026

Minh Nhat Nguyen, Dang Quang Pham, Thien Phuc Vo, Phuc Loc Dang, Duc Khoan Le
xdnhatnguyen.github.io/OrthoStyle (The 15th International Symposium on Information and Communication Technology (SOICT 2026) - Oral Presentation)

Awards

Quarter-finalist (Top 36 / 200+ teams) — Ho Chi Minh City, Vietnam, May 2026

Coding Challenge 2026, Faculty of Information Technology, University of Science, VNUHCM

Projects

Video Retrieval Pipeline — AI Challenge HCMC 2026 Aug 2026

Team lead building an end-to-end multimodal video retrieval pipeline addressing KIS, VQA, and temporal alignment using LanceDB hybrid search (SigLIP + BM25) and dynamic programming.

Latent Patch-Shuffle Style Transfer Aug 2026

Researched and implemented a training-free style transfer framework on latent diffusion models.

Handwritten Digits Classification Nov 2025

Led research benchmarking CNN vs. MLP architectures from scratch in PyTorch for the MNIST dataset.